

WATER CASE STUDY – MCQ NOTES (PART 1)

1. Topics Covered (VERY IMPORTANT – Direct MCQs)

- Water
- Air Pollution
- Solid Waste Management
- Industrial Pollution
- Man-made Disasters

👉 ⚠️ MCQ Trap: “Case studies include ONLY water” ❌ (Wrong — multiple topics)

2. Cape Town Water Crisis (CORE CASE STUDY)

Location

- City: **Cape Town**
 - Country: **South Africa**
 - Region: **Western Cape**
-

3. Key Event: “Day Zero”

- Refers to:
 - 👉 When **municipal water supply is shut down**
- People must collect water from **distribution points**

👉 ⚠️ MCQ Trap:

- Day Zero = water shortage ❌
 - Day Zero = **complete shutdown of piped water** ✅
-

4. Main Cause of Crisis (VERY IMPORTANT)

- Severe **drought (2015–2018)**
- **Low rainfall since 2015**
- Reservoir levels dropped **below 13.5%**

👉 If continued → **water supply shut down**



5. Climate of Cape Town (FREQUENT MCQ)

- Type: **Mediterranean Climate**
- Features:
 - Wet, cool winters
 - Dry, warm summers

👉 ⚠️ MCQ Trap:

- Rainfall season = **Winter (May–August)** ✓
 - Not summer ✗
-



6. Water Supply System

- Stored in **6 major reservoirs**
 - Sources:
 - Rainfall
 - Small groundwater contribution
 - Managed by:
 - 👉 **Western Cape Water Supply System (WCWSS)**
-



7. Water Distribution (IMPORTANT NUMBERS)

- **70%** → Urban + Industrial use
- **30%** → Agriculture

👉 ⚠️ Reverse values = common MCQ trap ✗



8. Why Shortages Happened (MULTIPLE FACTORS)

Primary Cause:

- Lack of rainfall

Additional Causes:

- High water consumption
 - Lack of investment in supply capacity
 - System sensitivity to rainfall changes
-

9. Population Growth Impact

- 1980 → **1.6 million**
- 2018 → **4+ million**

👉 Effect:

- Huge increase in water demand
-



10. Per Capita Water Use Change

- Early 1980s → **500,000 L/year per person**
- 2016 → **200,000 L/year per person**

👉 Means:

- Demand management reduced usage
-



11. Important Concept: Demand vs Supply

- Cape Town focused on:
👉 **Reducing demand instead of increasing supply**

👉 MCQ Insight:

- Demand reduction = **faster + cheaper** ✅
-



12. Virtual Water (VERY IMPORTANT CONCEPT)

Definition:

- Water used in producing goods that are **exported**

👉 Example:

- Food, textiles, products

👉 Key Fact:

- **South Africa = Net exporter of virtual water**
-

13. Role of Agriculture

- Agriculture consumes **large water share**
 - Increase in agricultural use → worsened shortages
-

14. Social Factors

- Formalization of informal settlements:
 - Shift from communal taps → individual usage
 - Slight increase in water demand
-

15. Crisis Management Actions (VERY IMPORTANT)

Government Actions:

- Water demand management
- New tariffs (penalty for excess use)
- Restricted water availability
- Installed water management devices

Public Actions:

- Behavior change (saving water)

Multi-level Involvement:

- Citizens

- Government (local + regional)
 - Private sector
 - Academic institutions
-

16. Key Lessons (HIGH MCQ PROBABILITY)

- Water shortages will **increase due to climate change**
 - Need:
 - Diversified water sources
 - Early warning systems
 - Effective communication
 - Cooperation across government levels
-

ULTRA-IMPORTANT ONE-LINERS (REVISION GOLD)

- Day Zero = **No piped water supply**
- Cape Town climate = **Mediterranean**
- Rainfall season = **Winter**
- Reservoir crisis level = **13.5%**
- WCWSS = **Water supply authority**
- 70% water → **Urban + Industrial**
- South Africa = **Virtual water exporter**
- Best solution = **Demand management**

WATER CASE STUDY – MCQ NOTES (PART 2)

(Cape Town: Management + Solutions + Impact)

1. Water Restrictions (VERY IMPORTANT)

Key Features:

- Increasingly **strict over time**
- Based on **severity of crisis**

Penalties:

- Fine: **R11,000 (~\$70) to R10,000 (~\$700)**
 - 👉 Used for:
- High consumption
- Rule violations

Extreme Restriction:

- **Level 6B**
 - Limit = **50 liters/person/day**

👉 ⚠️ MCQ TRAP:

- 50 liters = per household ❌
- 50 liters = **per person per day** ✅

Rules:

- Limits on:
 - Where water is used
 - Quantity
 - Purpose
- Revised:
 - 👉 **Monthly (ad-hoc basis)**
- Normally:
 - 👉 Requires **city council approval**

2. Tariff System (VERY IMPORTANT)

Type:

- **Step Tariff System**

👉 Meaning:

- More usage → **higher cost per unit**
-



Key Changes:

- Tariffs increased **significantly (2016–2018)**
 - Adjusted based on:
 - Restriction level
 - Sanitation charge:
 - 👉 Based on **% of water consumption**
-



Special Facts:

- Normally → **1 tariff change/year (July)**
 - Crisis → Multiple increases
-



Free Water Policy:

- Before drought:
 - **6000 L/month free (all households)**
 - +4500 L extra for **indigent households**
 - After crisis:
 - Free water removed for **non-indigent**
 - Fixed monthly charge introduced
-



IMPORTANT CONCEPT:

- Conflict:
 - 👉 Reduce water use vs maintain **utility profit**

👉 Global fact:

- Many countries' tariffs **don't cover costs (GLAAS report 2017)**
-



3. Water Management Devices (WMDs)

Definition:

- Installed in pipes to **limit daily water use**

Function:

- After limit → water flow becomes **trickle**
-

Limits:

- **350 L/day per household**
OR
 - **10,500 L/month**
-

Special Case:

- Applied to:
 - 👉 High-consumption households
 - 👉 Even after warnings
-

Origin:

- Initially for:
 - Debt control
 - Leak management in poor households
-

4. Pressure Reduction Strategy

Action:

- Reduced **water pressure in pipes (since 2017)**

Benefits:

- Lower consumption
- Reduced leaks
- Fewer pipe bursts

👉 ⚠️ MCQ Insight:

- Pressure reduction = **indirect demand control**
-

5. Communication Strategy (VERY IMPORTANT)

 **Timeline:**

- Started: **2016–2017**
-

 **Methods:**

- Websites → show:
 - Dam levels
 - Consumption
 - Media:
 - Radio ads
 - Bills
 - Flyers
 - Billboards
-

 **Strict Measures:**

- “Day Zero” warnings
 - Personalized warning letters
 - Public shaming:
 - 👉 Names of top “**water guzzlers**” published
 - Citizens encouraged to:
 - 👉 Report violators
-

 **Result:**

- Changed **social behavior + norms**
-



6. Water Consumption Breakdown

Largest Share:

- **Households (~53%)**

👉 Others:

- Retail & offices
- Industry
- Government
- Informal settlements

👉 ⚠️ MCQ: Domestic use = **highest** ✓

7. Smart Water Technologies

Smart Water Meters:

- Installed in **350+ schools**

Collaboration:

- Government + Private companies + Universities
-

Mobile Apps:

- **Tap Off**
 - Gamification (leaderboards)
- **Drop Drop**
 - Shows usage data
 - Helps track consumption

👉 Developed by:

- **University of Cape Town**
-

8. Sector-wise Reduction

Domestic:

- 69% of total use → primary focus



Targets:

- Domestic → **20% reduction**
 - Commercial/Industrial → **45% reduction**
-



Agriculture:

- Strict quota limits
-



9. Overall Impact (VERY IMPORTANT NUMBERS)



Consumption Drop:

- Feb 2015 → **1200 million L/day**
- Feb 2018 → **500 million L/day**



🔥 Huge reduction (~60%)



Household Behavior:

- <10,500 L/month users:
 - Increased from **64%** → **81%**
 - <6000 L/month:
 - Increased from **31%** → **49%**
-



10. Desalination (IMPORTANT CONCEPT)

Initial Plan:

- 9 plants + barge
 - ~**100 million L/day**
-

Reality:

- Not:
 - Cost-effective
 - Reliable
-

Final Output:

- Only **3 plants**
- **~16 million L/day**

👉 ⚠️ MCQ:

- Desalination = **minor contribution** ✅
-

11. Water Recycling

Before Crisis:

- Only **8% wastewater reused**
-

Plan:

- 50 million L/day → later **70 million L/day**

👉 Use:

- Irrigation
 - Industrial
 - Drinking (after treatment)
-

12. Spring Water Collection

Key Fact:

- Public collection points increased

Example:

- Visitors increased:

- 100 → **7000/month**
 - Some → **2000/day**
-

13. Policy Recommendations (VERY IMPORTANT)

- Diversify water supply
 - Long-term planning
 - Behavior change
 - Strong communication
 - Early warning systems
 - Government coordination
 - Climate change policies
 - Financial sustainability of utilities
-

ULTRA-IMPORTANT REVISION POINTS

- Level 6B = **50 L/person/day**
- Step tariff = **more use → higher price**
- WMD = **limits + trickle flow**
- Pressure reduction = **less leakage**
- Households = **highest water use (~53%)**
- Consumption reduced from **1200 → 500 ML/day**
- Desalination = **low impact**
- Apps = **Tap Off & Drop Drop**

PART 3: SOLID WASTE MANAGEMENT – PANAJI (MCQ NOTES)

1. Basic Facts (VERY IMPORTANT)

- City: **Panaji**

- State: **Goa (India)**
 - Population: **114,759 (Census 2011)**
 - Known for:
 - Tourism
 - Portuguese influence
-

2. Turning Point

- Landfill closed: **2005**
 - Shift to:
 - 👉 **Sustainable waste management**
-

3. Source Segregation (VERY IMPORTANT)

🔑 5 Waste Categories (COLOR CODED):

-  Green → **Wet waste**
-  Black/Grey → **Glass & metals**
-  Pink → **Paper & cartons**
-  Orange → **Plastics**
-  White → **Non-recyclables**

👉 ⚠️ MCQ Trap:

- Color mismatch questions VERY common
-

4. Collection System

- Door-to-door collection 
 - Wet waste → **Daily**
 - Dry waste → **Weekly**
-

5. Waste Processing

- Wet waste:
 - Composting (96 decentralized units)
 - Used for **horticulture**

- Commercial waste:
 - Window composting
- Hotels:
 - Biogas production

👉 Total processed:

- **~24 tons wet waste/day**
-



6. Dry Waste Handling

- Stored in:
 - 👉 **12 sorting units**
 - Further segregation:
 - 👉 **20 streams**
 - Sent to:
 - Cement plants (Wadi, Karnataka)
 - Recyclables → auctioned
-



7. Financial Management (IMPORTANT)

- Household fee:
 - 👉 **₹500/month**
 - Commercial:
 - 👉 **₹600 – ₹11,000**
 - Revenue:
 - 👉 **\$22,000 (2016)** from recyclables
-



8. Institutional Setup

- Managed by:
 - 👉 **City Solid Waste Management Department**
 - Features:
 - 24-hour helpline
 - Complaint redress system
 - Emergency vehicles
-

9. Success Factors

- Public engagement
 - Financial sustainability
 - Strong institutions
-

10. Awareness Programs

- Campaign: “**Bin Free in 2003**”
 - School program: “**Waste Wise**”
-

11. Pilot + Enforcement

- Pilot:
 - 70 households
 - 2 waste streams
 - Enforcement:
 - No community bins ❌
 - Door-to-door mandatory
 - Daily monitoring
-

PART 4: MARINE LITTER (VERY IMPORTANT MCQ SECTION)

1. Key Facts

- 62% of ocean debris = **Plastic**
- Plastics persist for:
 - 👉 **Decades to centuries**

2. Sources

- 80% from land-based sources

Examples:

- Mismanaged landfills
- Stormwater discharge
- Sewage
- Industrial waste
- Coastal tourism

3. Entry Points (5 Types)

- Rural areas (no collection)
- Poor urban infrastructure
- Overloaded cities
- Illegal dumping
- Open dump sites

4. Important Data (VERY IMPORTANT)

- 2010:
 - 👉 32 million tons mismanaged plastic
- Ocean leakage:
 - 👉 4.8 – 12.7 million tons

5. Plastic Value Concept

- Low-value plastic:
 - 👉 More leakage

👉 Reason:

- Waste pickers ignore it
-

6. Country Examples

- Philippines:
 - 👉 74% leakage = **uncollected waste**
 - China:
 - 👉 84% leakage = **uncollected waste**
-

7. Solutions (McKinsey – VERY IMPORTANT)

6 Key Solutions:

1. Increase collection
 2. Close leakage points
 3. Improve recycling systems
 4. Convert waste → fuel/electricity
 5. Increase recycling of low-value plastic
 6. Use plastic for industrial fuel
-

8. Great Pacific Garbage Patch

- Location:
 - 👉 **North Central Pacific Ocean**
 - Type:
 - 👉 **Ocean gyre accumulation zone**
 - Discovered:
 - 👉 **1988 (NOAA)**
-

PART 5: INDUSTRIAL DISASTER – BHOPAL GAS TRAGEDY

1. Basic Facts (VERY IMPORTANT)

- Date: **2–3 December 1984**
- Location: **Bhopal, India**

- Gas: **Methyl Isocyanate (MIC)**
-



2. Company

- **Union Carbide Corporation (UCC)**
 - Indian unit:
 - 👉 **UCIL**
-



3. Chemical Process

- MIC produced using:
 - Methylamine
 - Phosgene
 - Used to make:
 - 👉 **Carbaryl (pesticide)**
-



4. Disaster Cause (CORE)

- Water entered MIC tank (E610)
 - Caused:
 - 👉 Exothermic reaction
 - 👉 Temperature > **200°C**
-



5. Gas Leak

- MIC released:
 - 👉 **~30–40 tons**
 - Time:
 - 👉 **45–60 minutes**
-



6. System Failures (VERY IMPORTANT)

- Refrigeration system OFF ❌
- Flare tower NOT working ❌

- Gas scrubber inactive ✖
 - Alarm system delayed ✖
-

7. Gas Behavior

- Denser than air → stayed low
 - Spread:
 - 👉 Southeast direction
-

8. Other Toxic Gases

- Chloroform
 - Hydrogen chloride
 - Amines
-

9. Public Impact

- No warning system
 - People exposed directly
 - Many opened doors → increased exposure
-

10. Key Lessons

- Industrial safety critical
 - Need:
 - Emergency planning
 - Monitoring systems
 - Strict regulations
-

FINAL ULTRA-REVISION (ALL-IN-ONE)

 MOST ASKED MCQs:

- Level 6B → **50 L/day**
- Plastic in ocean → **62%**
- Land-based marine litter → **80%**
- Panaji landfill closed → **2005**
- MIC gas → **Bhopal disaster**
- Garbage patch → **Pacific gyre**
- Wet waste → **Green bin**
- Desalination → **low impact**

PART 6: BHOPAL GAS TRAGEDY – CONSEQUENCES & LEGAL (MCQ NOTES)

1. Immediate Health Effects (VERY IMPORTANT)

Symptoms:

- Coughing
- Severe eye irritation
- Suffocation
- Burning in respiratory tract
- Breathlessness
- Stomach pain

- Vomiting
-



2. Death Toll (VERY IMPORTANT NUMBERS)

Official:

- Immediate deaths → **2,259**
- Official confirmed (1991) → **3,928**
- Madhya Pradesh govt → **3,787**

Other Estimates:

- ~8,000 deaths in 2 weeks
- 8,000+ later deaths

👉 ⚠️ MCQ Trap:

- Multiple figures exist → question may ask **which is official**
-



3. Causes of Death

- Choking
 - Reflexogenic circulatory collapse
 - Pulmonary edema
-



4. Internal Damage (AUTOPSY FINDINGS)

- Lungs → severe damage
 - Brain → cerebral edema
 - Kidneys → tubular necrosis
 - Liver → fatty degeneration
 - Intestines → necrotizing enteritis
-



5. Impact on Children (VERY IMPORTANT)

- Stillbirth rate → ↑ **300%**
- Neonatal mortality → ↑ **200%**

Other Effects:

- Failure to grow
 - Development issues
-

6. Genetic & Long-term Effects

- Chromosomal abnormalities
 - Cancer
 - Immune deficiency
 - Neurological problems
 - PTSD
-

7. Long-Term Health Effects

Eyes:

- Chronic conjunctivitis
- Corneal scars
- Cataracts

Respiratory:

- Pulmonary fibrosis
 - Chronic bronchitis
 - TB aggravation
-

Neurological:

- Memory loss
 - Motor skill impairment
 - Numbness
-

Psychological:

- PTSD
-

8. Hidden / Late Effects

- Respiratory insufficiency
 - Cardiac issues
 - Cancer
 - Tuberculosis
-

9. Legal Issues (VERY IMPORTANT)

Company Responsibility

- Company: **Union Carbide Corporation (UCC)**
 - CEO: **Warren Anderson**
-

Key Legal Principle

Absolute Liability

Definition:

- If industry handles hazardous material →
 **Fully liable for damage**
 - No excuse allowed:
 - “No negligence” ❌
 - “Took precautions” ❌
-

Cases Filed

- Civil + Criminal cases in **Bhopal district court**
-

Major Issue

- Lack of accountability
 - Poor government response
-

Compensation Issues (VERY IMPORTANT)

Initial Claims:

- Govt estimate:
 - 3,000 deaths
 - 30,000 serious injuries
-

Reality:

- 573,588 people affected
 - ~70% of city population
-

Classification:

- Death cases → 5,295
 - Injured → 568,293
-

Compensation:

- Interim relief over years
 - Final settlement:
 - 👉 < ₹15,000 per victim (very low)
-

10. Key Observations (MCQ GOLD)

- Two tragedies:
 - Immediate gas leak
 - Long-term suffering
- System failures:

- Legal
 - Medical
 - Administrative
-



11. Ongoing Issues

- Toxic waste still present
 - Media raised:
 - 👉 **Accountability + liability questions**
-



FINAL ULTRA-REVISION (COMPLETE SYLLABUS)



TOP MCQ QUESTIONS:

- MIC gas = **Bhopal disaster**
 - Day Zero = **no piped water**
 - Plastic in ocean = **62%**
 - Marine litter land source = **80%**
 - Wet waste bin = **Green**
 - Absolute liability = **no defense allowed**
 - Stillbirth ↑ = **300%**
 - Neonatal mortality ↑ = **200%**
 - Desalination = **low impact solution**
 - Panaji landfill closed = **2005**
-

PART 7: BHOPAL-LIKE SITES IN INDIA (MCQ NOTES)

1. Key Concept

- Identified by:
 - 👉 Central Pollution Control Board (CPCB)
- Called:
 - 👉 “Bhopal-like disasters in the making”

👉 ⚠️ MCQ:

- Authority = CPCB 

2. Important Locations (VERY IMPORTANT LIST)

 Kerala

- **Eloor–Edayar (Cochin)**
 - **200,000 tons hazardous chemicals + pesticides**
-

● **Tamil Nadu**

- **Ranipet (Chromium contamination)**
 - **~ 220,000 tons waste**
 - **Covers 2–4 hectares**
-

● **Madhya Pradesh**

- **Ratlam Industrial Area**
 - **Pharma industry contamination**
-

● **Odisha**

- **Sukinda Valley**
 - **~ 50,000 tons chromium waste**
 - **Talcher**
 - **~ 60,000 tons waste**
 - **Ganjam**
 - **50,000 tons mercury waste**
-

● **Uttar Pradesh**

- **Jajmau–Kanpur**
 - **~ 100,000 tons hexavalent chromium**
 - **Rania (Kanpur Dehat)**
 - **~ 45,000 tons waste**
-

● **West Bengal**

- **Nivra Village**
 - **~ 4,440 tons chromium waste**
-

● **Lucknow**

- **Persistent Organic Pollutants (POP) site**
 - ~ 36,432 tons HCH waste
-

MCQ GOLD

- Chromium contamination = **most common**
 - Hazard types:
 - Chromium
 - Mercury
 - Pesticides
-

PART 8: GLOBAL FOOD WASTE INITIATIVE (VERY IMPORTANT)

1. Definitions (VERY IMPORTANT DIFFERENCE)

Food Loss:

- Occurs:
 - 👉 Production / storage / processing
 - Cause:
 - 👉 Unintentional
-

Food Waste:

- Occurs:
 - 👉 Consumer level
 - Cause:
 - 👉 Intentional discard
-

👉 ⚠️ **MCQ Trap:**

- Loss = unintentional 

- Waste = intentional 
-



2. Where Loss Occurs (SUPPLY CHAIN)

Stages:

1. Production
 2. Handling & Storage
 3. Processing & Packaging
 4. Distribution & Marketing
 5. Consumption
-



3. Distribution of Food Loss (VERY IMPORTANT NUMBERS)

- Production → **30%**
- Handling & storage → **21%**
- Processing → **4%**
- Distribution → **15%**
- Consumption → **30%**

👉 ⚠️ MCQ:

- Highest = **Production + Consumption (equal)**
-



4. Global Data (VERY IMPORTANT)

- **1/3rd food wasted globally**
 - Economic loss:
 - 👉 **\$940 billion/year**
 - FAO:
 - 👉 **32% food lost (2009)**
 - Calories:
 - 👉 **24% wasted (~1 in 4 calories)**
-



5. Environmental Impact

- Greenhouse gas emissions
 - Water wastage
 - Land misuse
 - Ecosystem damage
-



6. Social Impact

- Saving 25% food →
👉 Feed **870 million people**
-



7. Initiatives (IMPORTANT)

- **SAVE FOOD** (FAO + Messe Düsseldorf)
 - **Think.Eat.Save** campaign
 - **Global FoodBanking Network**
 - **OECD Food Chain Network**
 - **FUSIONS** (EU program)
-



8. Commodity Waste (VERY IMPORTANT)

- Cereals → **Highest share (~53% kcal)**
 - Fruits & vegetables → High losses (by weight)
-



9. Regional Trends

- Developed countries:
👉 Waste at **consumer level**
 - Developing countries:
👉 Loss at **production stage**
-



10. Causes of Food Waste

Direct Causes:

- Poor quality
 - Aesthetic rejection
 - Expiry dates
 - Overproduction
-

Indirect Causes (VERY IMPORTANT TYPES):

1. Technological → poor storage
 2. Managerial → poor planning
 3. Behavioral → attitudes
 4. Structural → policies, climate
-

11. Key Insight (MCQ TRAP)

- As economy develops:
 - 👉 Loss ↓
 - 👉 Waste ↑
-

12. SDG Connection

- Helps achieve:
 - SDG 2 (Zero Hunger)
 - SDG 12 (Responsible Consumption)
 - SDG 13 (Climate Action)
-

FINAL SUPER REVISION (ENTIRE SYLLABUS)

MOST IMPORTANT MCQs:

- CPCB → Hazard sites authority
- Food loss = **production stage**
- Food waste = **consumer stage**
- 1/3rd food wasted globally
- \$940 billion loss

- Highest waste stages = **production & consumption**
 - Chromium contamination → most common toxic site
 - 870 million people can be fed
-

PART 9: UNDERLYING DRIVERS OF FOOD LOSS & WASTE (MCQ NOTES)

1. Core Concept (VERY IMPORTANT)

- Food loss/waste happens due to:
 - 👉 **Direct causes + Underlying drivers**
-

2. Direct Causes (MCQ BASIC)

- Deterioration
 - Suboptimal quality
 - Appearance issues
 - No buyer / demand
-

3. Types of Underlying Drivers (VERY IMPORTANT CLASSIFICATION)

1. Technological Drivers

- Poor infrastructure

- Inadequate equipment
- Suboptimal packaging

👉 Example:

- No cold storage → food spoilage
-



2. Managerial Drivers

- Poor management skills
 - Inflexible procurement rules
 - Poor supply-demand forecasting
 - Inefficient marketing
-



3. Behavioral Drivers

- Norms & attitudes
- Lack of awareness
- Risk concerns (food safety fear)

👉 Example:

- Rejecting “ugly” fruits
-



4. Structural Drivers

- Demographics
 - Climate conditions
 - Policies & regulations
 - Economics
 - Access to financing
-



4. Key Insight (VERY IMPORTANT)

- Drivers are:
 - 👉 **Interconnected**

👉 Example:

- Poor storage → due to lack of money

5. Regional Difference (MCQ TRAP)

- Low-income countries:
 - 👉 Loss due to **infrastructure issues**
 - High-income countries:
 - 👉 Waste due to **consumer behavior**
-

6. Development Trend (VERY IMPORTANT)

- Economic development leads to:
 - Loss ↓
 - Waste ↑
-

PART 10: SOLUTION FRAMEWORK (VERY IMPORTANT)

1. Target–Measure–Act (CORE FRAMEWORK)

Step 1: Target

- Set reduction goals

Step 2: Measure

- Track where waste occurs

Step 3: Act

- Implement solutions

👉 ⚠️ MCQ:

- Sequence = **Target** → **Measure** → **Act** ✓

2. Key Strategy Insights

- No single solution ❌
- Multiple interventions needed ✅
- Collaboration is critical

3. Important Principles

- Awareness = starting point
- Business case needed
- Avoid “knock-on effects”
- Country-specific solutions

PART 11: ACTORS & THEIR ROLES (VERY IMPORTANT MATCHING MCQs)

1. Farmers

- Improve harvesting techniques
- Use better tools
- Access markets

2. Fishers

- Reduce bycatch
- Improve storage

3. Animal Farmers

- Reduce contamination
 - Improve handling
-



4. Processors

- Improve processing efficiency
 - Reduce errors
 - Improve packaging
-



5. Packaging Providers

- Better packaging design
 - Reduce damage
-



6. Transport & Logistics

- Cold chain
 - Better transport timing
-



7. Retailers

- Better inventory management
 - Flexible standards
 - Avoid overstock
-



8. Households (VERY IMPORTANT)

- Buy only needed food
 - Proper storage
 - Use leftovers
-



9. Restaurants

- Reduce portion size
 - Reuse excess food
 - Track waste
-



10. Hotels

- Buffet control
 - Donate extra food
-



11. Policymakers

- Improve infrastructure
 - Regulations
 - Awareness programs
-



12. Financiers

- Fund waste reduction tech
-



13. Researchers

- Develop new technologies
-



14. Civil Society

- Awareness campaigns
-

FINAL ULTRA-REVISION (COMPLETE COURSE)

TOP 15 MOST IMPORTANT MCQs:

1. Day Zero = **no piped water**
 2. Level 6B = **50 L/person/day**
 3. Plastic in ocean = **62%**
 4. Marine litter land source = **80%**
 5. Food wasted globally = **1/3rd**
 6. Economic loss = **\$940 billion**
 7. Highest waste stage = **production & consumption**
 8. Panaji landfill closed = **2005**
 9. Wet waste bin = **green**
 10. MIC gas = **Bhopal disaster**
 11. Stillbirth increase = **300%**
 12. Absolute liability = **no defense**
 13. CPCB = **hazard sites authority**
 14. Target-Measure-Act = **solution framework**
 15. Developed countries = **consumer waste**
-

PART 10: FINAL MCQ NOTES (LAST SECTION)

1. Role of Women in Reducing Food Loss & Waste (VERY IMPORTANT)

Key Facts:

- Women interact with food at:
 - 👉 Every stage (farm → fork)
-

Important Numbers:

- 41% of global agricultural workforce = women
 - Majority in:
 - 👉 South Asia & Sub-Saharan Africa
-
- Women responsible for:
 - 👉 85–90% household food preparation time

MCQ Insight:

- Targeting women =
👉 **More effective than general campaigns**
-

Examples:

- **Tanzania Initiative:**
 - Women farmers given:
 - Market access
 - Solar drying tech

👉 Result:

- Reduced fruit wastage
-

- **Australia Campaign (“1 Million Women”)**
 - Awareness campaign
 - Celebrity chef involvement
-

2. Case Study: SecondBite (Australia)

Organization:

- Non-profit:
👉 **SecondBite**
-

Function:

- Acts as:
👉 **Food redistribution broker**

Process:

- Collect surplus food
 - Distribute to:
 - 👉 Community groups + food banks
-



Collaboration:

- Works with:
 - Farmers
 - Retailers
 - Governments
-



Policy:

- Helped introduce:
 - 👉 **Good Samaritan Acts** (encourage donation)
-



Key Data:

- 2012:
 - 👉 3,000 metric tons food saved
-



MCQ Trap:

- SecondBite = waste reduction via:
 - 👉 **Redistribution**, not recycling ❌
-

3. Hydrothermal Carbonization (HTC) – Waste to Fuel

1. Definition (VERY IMPORTANT)

- Converts:
👉 **Wet biomass** → fuel (hydrochar)
-

2. Input Materials

- Food waste
 - Yard waste
 - Organic municipal waste
-

3. Outputs

- Hydrochar (fuel)
 - Bio-coal
 - Gas
 - Liquids
-

4. Key Advantage (VERY IMPORTANT)

- Works on:
👉 **Wet biomass (NO drying needed)**

👉 ⚠️ MCQ:

- Compared to incineration:
 - Drying required ❌
 - HTC avoids drying ✅
-

5. Process Conditions

- Temperature:
👉 ~ **180–250°C**
-

6. Applications

- Fuel pellets
 - Carbon sequestration (biochar)
 - Soil amendment
 - Adsorbents
-

7. Indian Example (VERY IMPORTANT)

- Developed by:
👉 IIT Kharagpur
-

8. Concept

- “Zero Loss Process”
 - Converts:
 - 👉 Municipal solid waste →
 - Biofuel
 - Fertilizer
 - Energy
-

MCQ GOLD:

- HTC = **wet waste conversion**
 - Product = **Hydrochar**
 - No drying needed 
-

FINAL SUPER REVISION (LAST ADDITIONS)

MOST IMPORTANT FROM THIS PART:

- Women = **41% workforce**
- Women = **85–90% food preparation**

- Best strategy = **target women**
- SecondBite = **food redistribution**
- 3,000 tons food saved (2012)
- HTC = **wet biomass** → **fuel**
- Temp = **180–250°C**
- IIT Kharagpur → HTC innovation